ME2605 · FALL 2026 · COURSE PROJECT

Project Intro

Instructor: Ruoyu Sun

Some types of projects

Examples:

A. Improving the training of a certain task — recommended

Take an open-source training task. It can be pretraining or post-training, on language, vision, robots, AI for science, etc.

Try tricks you learned from class or outside class. Explain your tricks.

Cut the time to the baseline's final loss by at least 10%.

(Just for example, a reasonable improvement can be more than 10%; you can judge yourself.)

Note: AI can modify code very easily.
You shall put some thought, and explain your effort in the final demo.

B. Mechanism

Explain why something works, or when it helps, from the lectures.
For example: the condition number and normalization.

C. Leaderboard — advanced

Submit a solution to a public training leaderboard.
Shall be among the top competiters.

Examples of mechanism studies

Does a smaller condition number train faster?

Lec 3–4 · two nets that differ only in κ

Is warmup still needed with normalized layers?

Lec 4, 7 · remove it under BN, LN, and no norm

Is Adam's advantage over SGD just conditioning?

Lec 4, 7 · shrink κ, see if the gap closes

Does the edge of chaos predict which nets train?

Lec 2 · sweep init variance against depth

Does a recipe tuned at width 256 survive 1024?

Lec 5 · the honest test of a speedrun

Ask AI for more ideas.

Then check them against the lectures, and pick one that the course already has the tools for.

Other benchmarks

Small speedruns: **Modded nanoGPT**, train a small GPT to a given validation loss of 3.28.
Larger-scale leaderboards: **MLPerf Training**; **PostTrainBench**.

Category	Benchmark	What it is	8×H100 time
Small	**modded-nanogpt**	124M GPT on FineWeb to a fixed val loss (3.28); architecture and data fixed by the rules	40 s (best record) – 45 min (baseline)
Small	**BabyLM**	Sample-efficient pretraining on developmentally plausible data, ≤10 epochs; Small = 10M words, Strict = 100M words	Small: ~1–2 min; Strict: ~5–10 min optimized, up to ~20 min plain HF
Large	**PostTrainBench**	Agents autonomously post-train 4 small base LLMs (1.7B–4B); bounded compute	10 h budget (wall-clock, including agent iterations, not pure training)
Large	**MLPerf Training**	Industry-standard time-to-quality suite across LLM workloads	GPT-OSS 20B ~3 h; Llama 8B ~4.6 h (estimated from B200)

Possible Tricks and Methods to Try in Your Project

Optimizer and step size

Schedule: warmup length, cosine or WSD decay, peak learning rate
Muon or Shampoo instead of AdamW on the matrix parameters
Gradient clipping, and loss scaling under fp8

The computation

FlashAttention
torch.compile, and fewer host–device synchronizations
Low precision training

Model and initialization

Normalization placement, residual scaling, init variance (Lec 2–4)
Width and depth, and the batch size they were tuned at (Lec 5)
Attention variant: GQA, RoPE base, tied embeddings

Some are taught later. You can also find other tricks online.

Migration of Training Tricks/Methods to NPU

Many public benchmarks **are heavily optimized already.**

Choice 1: Focus on **Migration to NPU.**

Yupeng Chen et al. have set up a leaderboard **for NPU**.

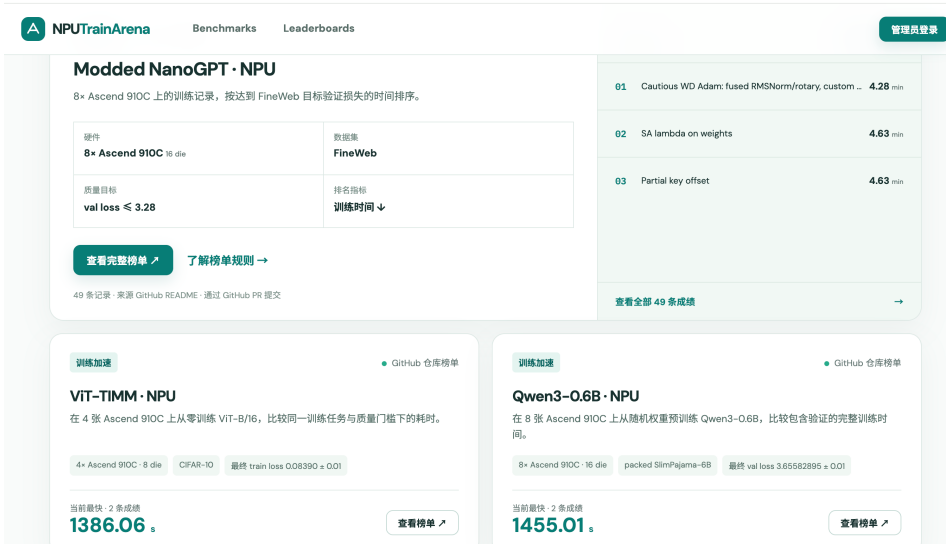
NPUArena
121.15.210.202:10001

The paper
Chen et al., arXiv:2610.29171

Tasks

modded-nanogpt-npu
gitcode.com/SLAI/modded-nanogpt-npu

nanovLM, AirBench, etc.



Training Models from your chosen field

Many public benchmarks **are heavily optimized already.**

Choice 2: consider training a frontier model from your own field instead.

- A paper from 2024 or later, or one with many citations.
- Take its model and its training recipe.
- Change the recipe; keep the model and the data.

An example: AI4Science

A single-cell foundation model, trained on 1.6M cells.
Build a smaller version: a subset of the datasets, and a narrower model of the same architecture.

Compute Resources

We will apply for Ascend cards.

- The target is 4× Ascend 910C.
- If that is delayed, we aim for at least one 910C, or one RTX 5090.
- The NPU leaderboard requires an Ascend 910C card.
- Hetao students already have accounts.

  Everyone else: we will try to open the accounts for you.

Write down the hardware you plan to use in the proposal.
If you are not sure yet, say so, and we will try to accommodate.

Timeline

Wed, Nov 18, 8pm	1–2 page proposal, to a submission website (shown later in the WeChat group). **Not graded; just for the record.** **No submission: subtract 5 points.**
After that	Generally, no major topic changes; if so, let the instructor know. Minor adjustments are fine.
Week 14	Submit a project report, 10–20 pages. (Tentative) poster session in a room. **Poster:** you shall be able to explain, at least roughly, what AI does and why it works. Don't say: I asked AI to improve the performance, and I don't know what it does.

Remark: do not hand in what AI can finish in an afternoon.

The grading depends on how much human effort has been put into a project.

Individual work: no teams, one project per person.

Proposal: what to write

One or two pages. You may consider the following points:

1 **The task and the baseline**

Repo link, the hardware you will use, and the baseline time-to-loss you will beat.

2 **What tricks or methods you intend to try**

3 **What compute you need**

Card type, number of cards, and hours per week.

4 **The risk**

What you do if the card does not arrive, or the first idea fails.